

Forever loops

Some loops run forever. If you set the condition in a “while” loop to be “True”, it can never be false and the loop will never end. This can either be useful or very annoying.

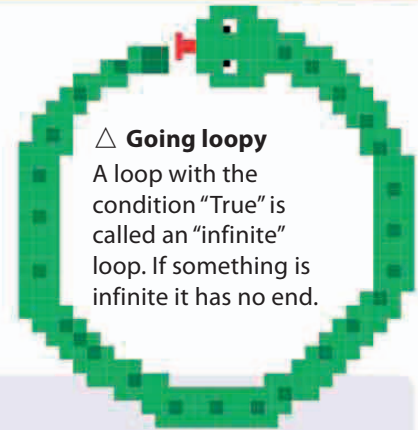
1 Create a forever loop

The loop condition here is set to “True”. Nothing that happens inside the loop will make “True” equal anything but “True”, so the loop runs forever.

The typed word is stored in the variable “answer”

```
while True:
    answer = input('Type a word and press enter: ')
    print('Please do not type \'\' + answer + \'\' again.')
```

The loop is always “True” so will never end



2 What the program looks like

On the opposite page the monster program’s loop condition checked to see what the user’s answer was. If the answer isn’t “y”, the loop will stop. The loop shown above doesn’t check the answer, so the user can’t make it stop.

```
>>>
```

```
Type a word and press enter: tree
Please do not type 'tree' again
Type a word and press enter: hippo
Please do not type 'hippo' again
Type a word and press enter: water
Please do not type 'water': again
Type a word and press enter
```

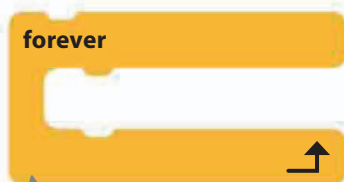
No matter what is typed, this loop just keeps on going



REMEMBER

“forever” block

Remember the “forever” block in Scratch? It repeats the code inside it until the red stop button is clicked. A “while True” loop does exactly the same thing. It can be used to make a program keep doing something, such as asking questions or printing a number, as long as the program is running.



The “forever” block keeps the sprite moving endlessly

EXPERT TIPS

Stop the loop

If you get stuck in an infinite loop, you can stop it from IDLE. Click in the Python shell window, then hold down the “CTRL” key and press the “C” key. This asks IDLE to stop the program. You might have to press “CTRL-C” a few times. This is similar to clicking the red stop button in Scratch.

